

SOFTWARE ENGINEERING LAB 2

Agile Backlog Creation & Sprint Simulation in Jira

Name	Ullas T	Section	F
SRN	PES1UG24AM309	Project	Faculty Research Grant & Publication Tracker

Problem Statement

Faculty Research Grant & Publication Tracker — this report documents the Agile backlog, priorities, story-point estimates, sprint simulation, burndown charts, and reflection for the Jira project.

1. Epics

Epic 1: Grant Fund Management

Manages grant fund allocations, grant expenses, and calculation of the remaining grant balance.

Epic 2: Publication & Citation Management

Manages faculty publications, publication indexing status, and citation metrics.

Epic 3: Co-author Approval Workflow

Supports submission of co-author details and review/approval by the Research Dean.

Epic 4: Grant Utilization & Analytics

Provides grant utilization status and fund burn-up analytics over time.

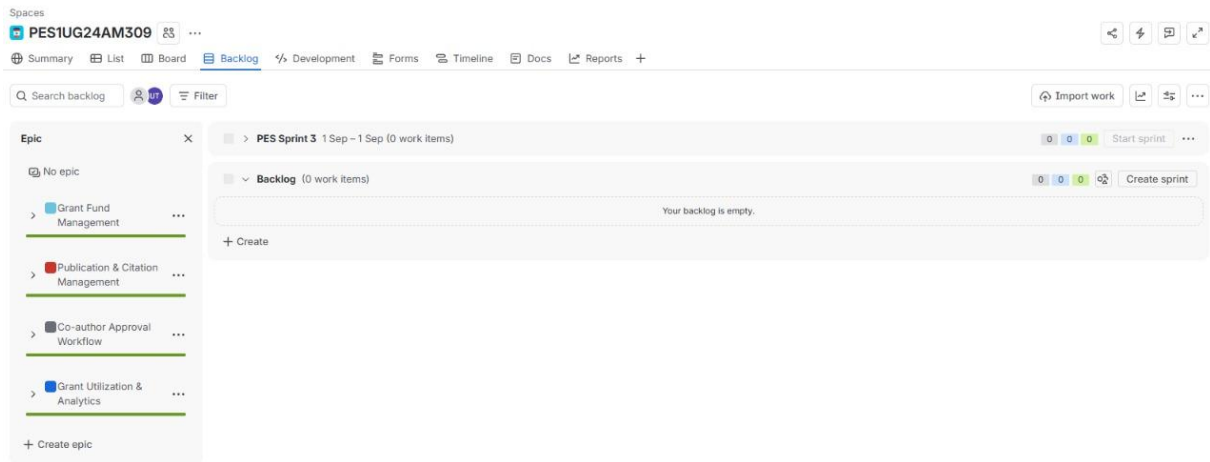


Figure 1: Jira backlog showing the four Epics.

2. User Stories

Epic 1: Grant Fund Management

Story	As a	I want	So that
US-1: Track Grant Fund Allocation	Faculty Researcher	To track grant fund allocations	I can monitor the budget allocated to the research grant.

Story	As a	I want	So that
US-2: Record Grant Expenses	Faculty Researcher	To record equipment procurement and other grant expenses	research expenditure is accurately tracked against the grant.
US-3: Calculate Remaining Grant Balance	Faculty Researcher	To calculate the remaining grant balance after expenses	I can know how much of the grant budget is still available.

Epic 2: Publication & Citation Management

Story	As a	I want	So that
US-4: Record Faculty Publications	Faculty Researcher	To record faculty publication details	I can maintain an accurate record of my research publications.
US-5: Track Publication Indexing Status	Faculty Researcher	To track the indexing status of publications	I can monitor the indexing status of my publications.
US-6: Track Citation Metrics	Faculty Researcher	To track citation metrics for publications	I can monitor publication performance and impact.

Epic 3: Co-author Approval Workflow

Story	As a	I want	So that
US-7: Submit Co-author Details	Faculty Researcher	To submit co-author details for approval	the required co-author information can be formally reviewed.
US-8: Review Co-author Approval	Research Dean	To review and approve submitted co-author details	co-author information is properly authorized before approval.

Epic 4: Grant Utilization & Analytics

Story	As a	I want	So that
US-9: View Grant Utilization	Faculty Researcher / Research Dean	To view grant allocation, expenses, and utilization status	stakeholders can monitor how sponsored research funds are being used.
US-10: Generate Fund Burn-up Analytics	Research Dean	To generate fund burn-up analytics showing expenditure against the allocated budget over time	I can monitor spending trends and remaining grant funds.

3. Priorities and Story Points

The User Stories were prioritized as High or Medium and estimated using Fibonacci-style story points, consistent with the lab handout.

Story	Priority	Story Points
US-1 Track Grant Fund Allocation	High	3

Story	Priority	Story Points
US-2 Record Grant Expenses	High	5
US-3 Calculate Remaining Grant Balance	High	3
US-4 Record Faculty Publications	High	3
US-5 Track Publication Indexing Status	High	3
US-6 Track Citation Metrics	High	5
US-7 Submit Co-author Details	High	3
US-8 Review Co-author Approval	High	3
US-9 View Grant Utilization	Medium	3
US-10 Generate Fund Burn-up Analytics	Medium	5

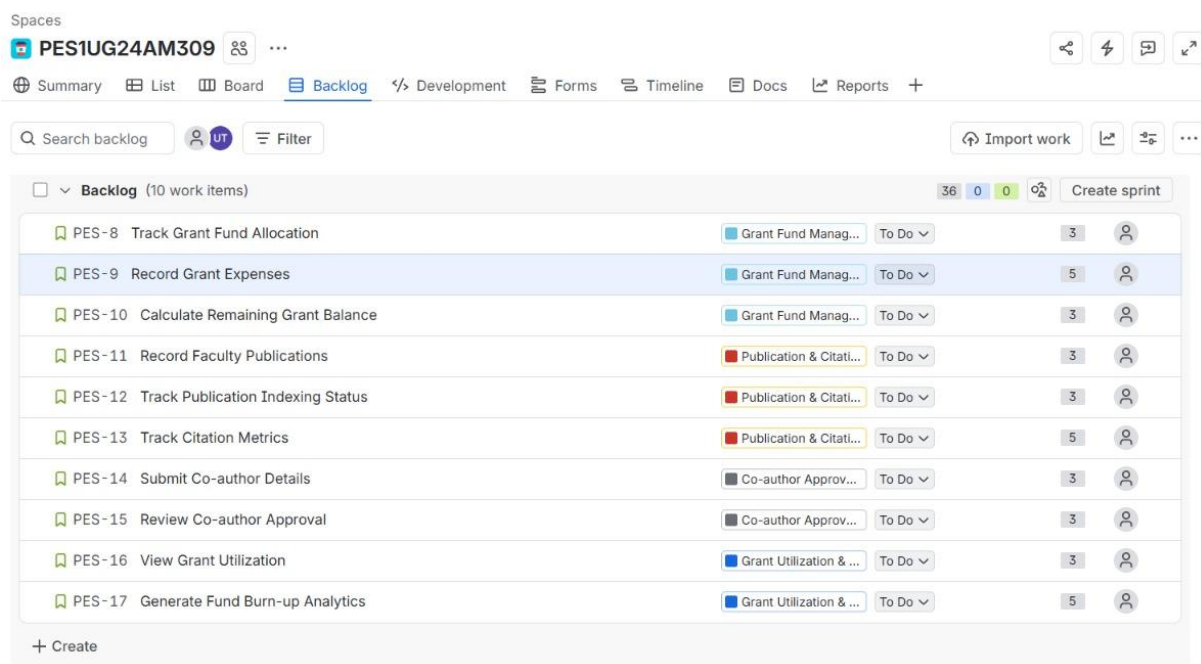


Figure 2: Initial Jira backlog with the ten User Stories.

4. Sprint Information

Sprint 1 (1 hour) — 5 stories, 17 story points

- Track Grant Fund Allocation — 3
- Record Grant Expenses — 5
- Calculate Remaining Grant Balance — 3
- Record Faculty Publications — 3
- Track Publication Indexing Status — 3

Sprint 2 (1 hour) — 5 stories, 19 story points

- Track Citation Metrics — 5
- Submit Co-author Details — 3
- Review Co-author Approval — 3

- View Grant Utilization — 3
- Generate Fund Burn-up Analytics — 5

The sprint simulation followed the Jira workflow: To Do → In Progress → In Review → Done.

5. Sprint Simulation

Sprint 1 — Board Progress

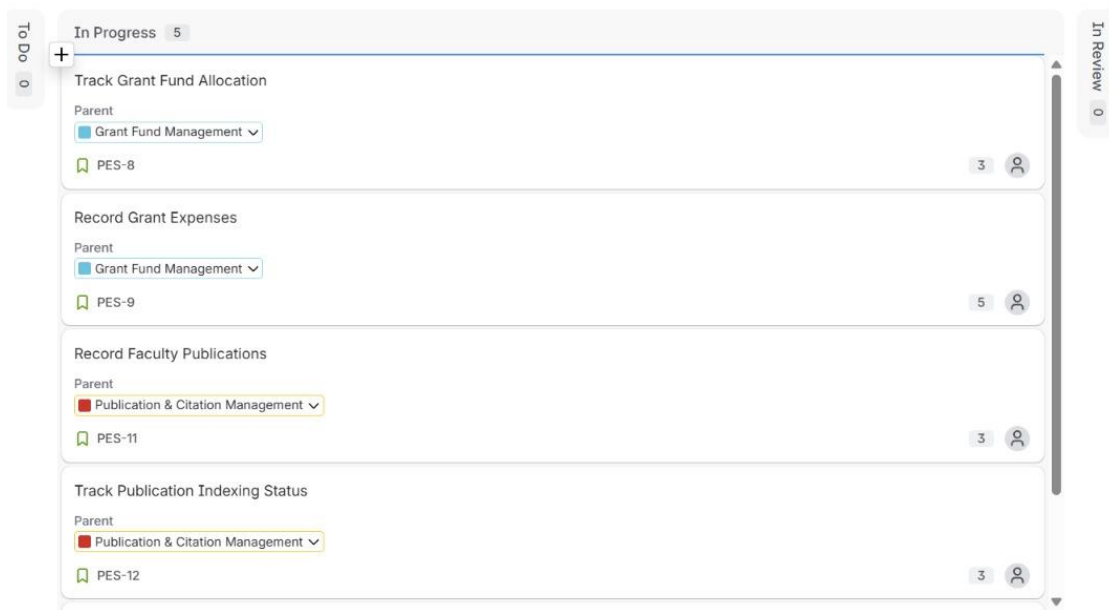


Figure 3: Sprint 1 stories moved to In Progress.

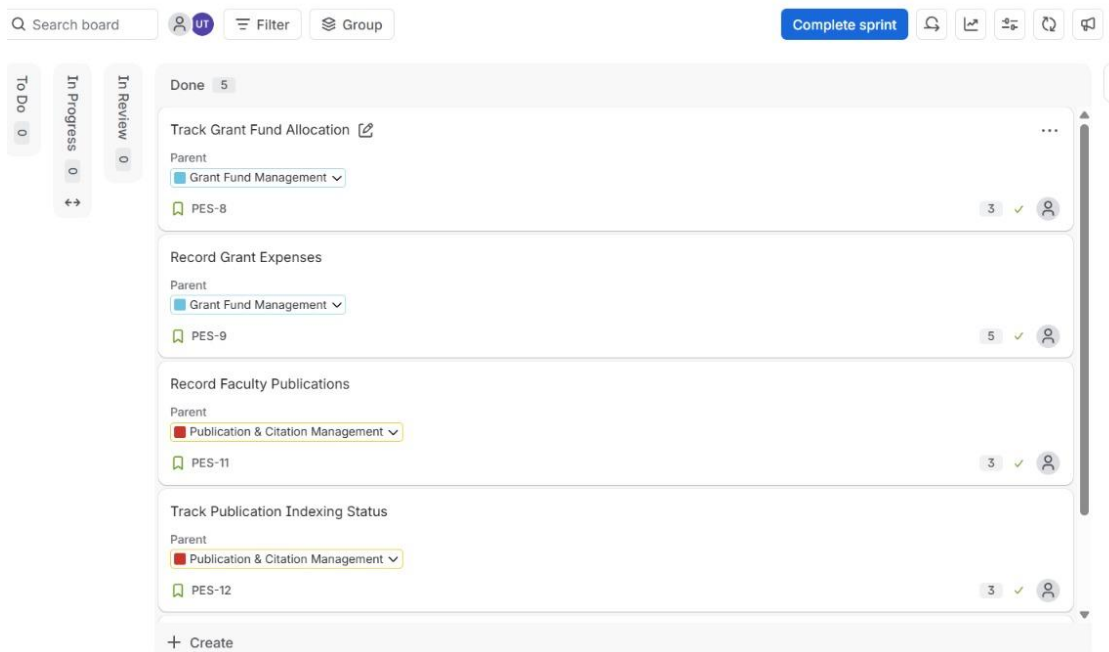


Figure 4: Sprint 1 stories shown in Done.

Sprint 2 — Board Progress

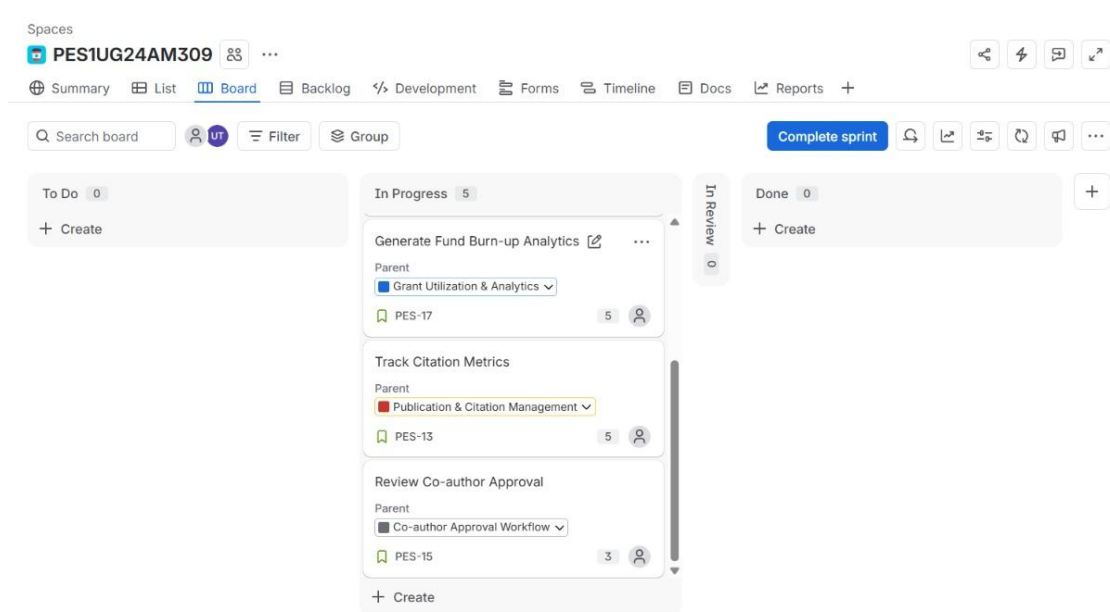


Figure 5: Sprint 2 stories moved to In Progress.

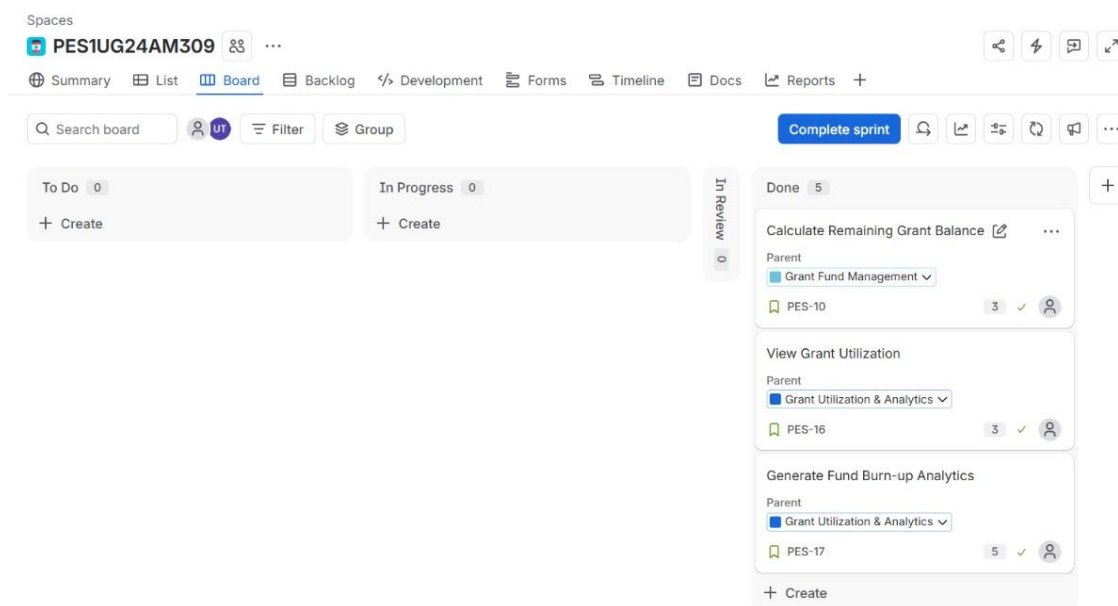


Figure 6: Sprint 2 stories shown in Done.

Final Backlog Evidence

Spaces

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Summary List Board Backlog Development Forms Timeline Docs Reports +

Search work Filter Group

	Work	Assignee	Reporter	Priority	Status	Resolution	Created	Updated	
☐	PES-4 Grant Fund Management	Unassigned	Ullas T	Medium	To Do	Unresolved	01 Sept 2026, 11:15	01 Sept 2026, 11:15	No
☐	PES-6 Track Grant Fund Allocation	Unassigned	Ullas T	High	Done	Done	01 Sept 2026, 11:21	01 Sept 2026, 12:26	No
☐	PES-9 Record Grant Expenses	Unassigned	Ullas T	High	Done	Done	01 Sept 2026, 11:25	01 Sept 2026, 12:26	No
☐	PES-10 Calculate Remaining Grant Balance	Unassigned	Ullas T	High	Done	Done	01 Sept 2026, 11:26	01 Sept 2026, 13:43	No
☐	PES-5 Publication & Citation Management	Unassigned	Ullas T	Medium	To Do	Unresolved	01 Sept 2026, 11:16	01 Sept 2026, 11:16	No
☐	PES-11 Record Faculty Publications	Unassigned	Ullas T	High	Done	Done	01 Sept 2026, 11:28	01 Sept 2026, 12:27	No
☐	PES-12 Track Publication Indexing Status	Unassigned	Ullas T	High	Done	Done	01 Sept 2026, 11:28	01 Sept 2026, 12:27	No
☐	PES-13 Track Citation Metrics	Unassigned	Ullas T	High	Done	Done	01 Sept 2026, 11:29	01 Sept 2026, 13:43	No
☐	PES-8 Co-author Approval Workflow	Unassigned	Ullas T	Medium	To Do	Unresolved	01 Sept 2026, 11:17	01 Sept 2026, 11:17	No
☐	PES-14 Submit Co-author Details	Unassigned	Ullas T	High	Done	Done	01 Sept 2026, 11:30	01 Sept 2026, 12:27	No
☐	PES-15 Review Co-author Approval	Unassigned	Ullas T	High	Done	Done	01 Sept 2026, 11:31	01 Sept 2026, 13:43	No
☐	PES-7 Grant Utilization & Analytics	Unassigned	Ullas T	Medium	To Do	Unresolved	01 Sept 2026, 11:18	01 Sept 2026, 11:18	No
☐	PES-16 View Grant Utilization	Unassigned	Ullas T	Medium	Done	Done	01 Sept 2026, 11:32	01 Sept 2026, 13:43	No
☐	PES-17 Generate Fund Burn-up Analytics	Unassigned	Ullas T	Medium	Done	Done	01 Sept 2026, 11:32	01 Sept 2026, 13:43	No

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4 of 4

Figure 7: Final Jira list showing priorities, status, and story point estimates.

Spaces

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Summary List Board Backlog Development Forms Timeline Docs Reports +

Search backlog Filter Import work

Backlog (10 work items) 0 0 0 Create sprint

PES-8	Track Grant Fund Allocation	To Do	-	
PES-9	Record Grant Expenses	To Do	-	
PES-10	Calculate Remaining Grant Balance	To Do	-	
PES-11	Record Faculty Publications	To Do	-	
PES-12	Track Publication Indexing Status	To Do	-	
PES-13	Track Citation Metrics	To Do	-	
PES-14	Submit Co-author Details	To Do	-	
PES-15	Review Co-author Approval	To Do	-	
☐ PES-16	View Grant Utilization	To Do	-	...
PES-17	Generate Fund Burn-up Analytics	To Do	-	

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Figure 8: Jira backlog showing all ten User Stories and estimates.

6. Burndown Charts

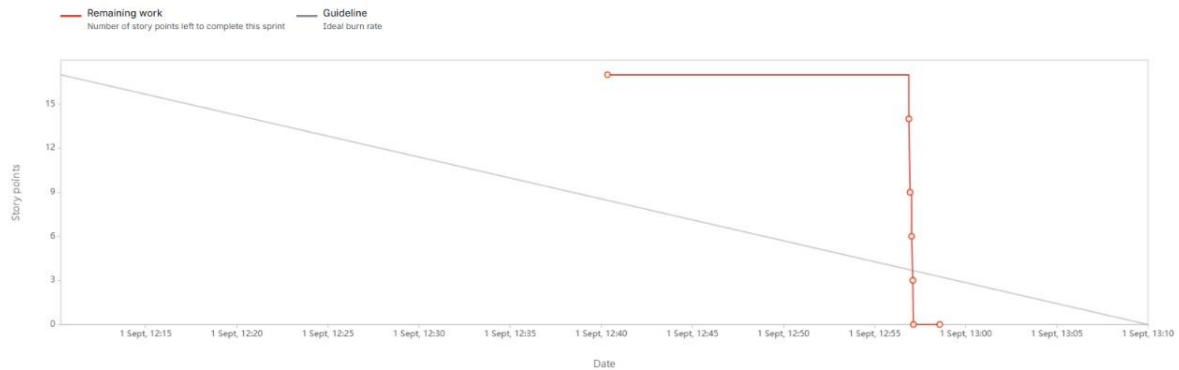
Sprint 1 Burndown Chart

Sprint burndown chart

[How to read this report](#)

Sprint: Estimation field:

Date - 1 September 2026 - 1 September 2026



Report: PES Sprint 1

*Work item added after sprint start

Figure 9: Sprint 1 Burndown Chart.

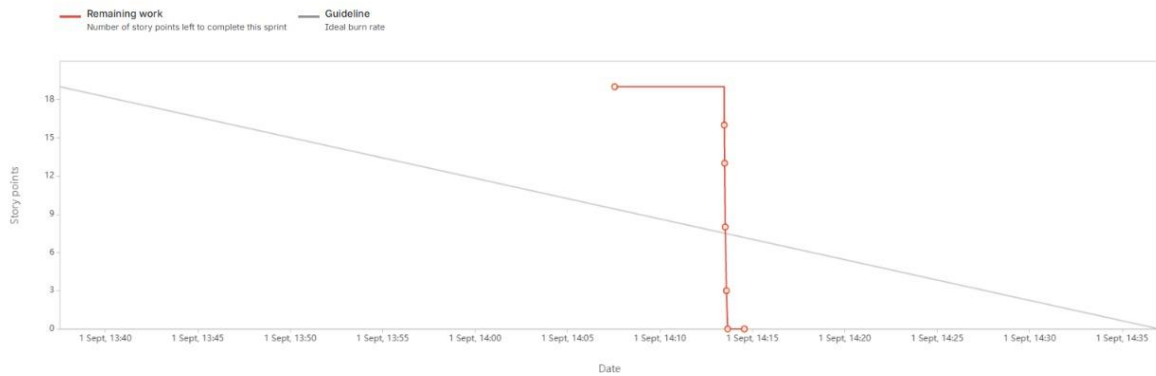
Sprint 2 Burndown Chart

Sprint burndown chart

[How to read this report](#)

Sprint: Estimation field:

Date - 1 September 2026 - 1 September 2026



Report: PES Sprint 2

*Work item added after sprint start

Figure 10: Sprint 2 Burndown Chart.

7. Reflection Questions

1. Did your estimations reflect the actual effort?

Yes. The story point estimates provided a reasonable relative representation of effort and complexity. Smaller activities were assigned 3 points, while comparatively more complex activities were assigned 5 points.

2. Was your backlog well-prioritized?

Yes. The backlog was prioritized according to user impact and business importance. Core grant and publication activities were given High priority, while grant utilization and analytics were assigned Medium priority.

3. How did your simulated sprint align with your plan?

The simulated sprints aligned with the plan. The ten User Stories were divided into two one-hour sprints, with 17 story points in Sprint 1 and 19 story points in Sprint 2. The stories were moved through the Jira workflow and completed.

4. What insights did the burndown chart give about your team's capacity?

The burndown charts showed the reduction in remaining story points during the simulated sprints. They provided a visual way to evaluate progress and the amount of work completed within the selected sprint duration.

Conclusion

The Jira exercise demonstrated backlog creation, prioritization, story-point estimation, sprint execution, workflow progression, and burndown-based progress analysis for the Faculty Research Grant & Publication Tracker.